MIL-P-15554B(SHIPS)
9 November 1961
SUPERSEDING
MIL-P-15554A
30 August 1952

MILITARY SPECIFICATION

PROPELLER REVOLUTION INDICATING EQUIPMENT, SYNCHRO

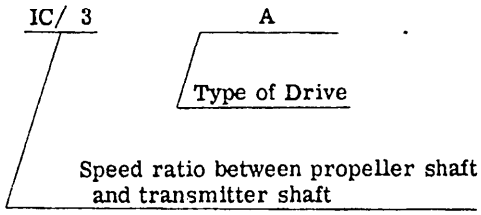
(NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope.- This specification covers instantaneous, continuously indicating synchro electrical propeller revolution indicating equipment consisting of transmitters, indicator-transmitters, and indicators for use in Naval shipboard interior communication system, circuit "K".

1.2 Classification.-

1.2.1 Transmitters.- The transmitters shall be of types designated in the following form, as specified (see 6.1).



The type designations shall be in accordance with the following symbols:

Basic indicator:

IC/ - Basic indicator for Naval interior communication equipment. Speed ratio between propeller shaft and transmitter shaft:

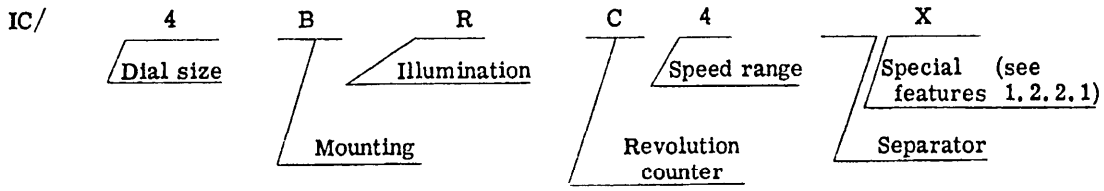
1 - 1 to 1
2 - 2 to 1
3 - 3 to 1

Type of drive: (Method of driving the transmitter from the propeller shaft or propulsion equipment).

A - Gear drive
B - Direct coupling

Operating speed range: 0-400 r.p.m. unless otherwise specified. (see 6.1).

1.2.2 Indicator-transmitters and indicators.- The indicator shall be of types designated in the following form, as specified (see 6.1):



FSC 6320

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The type designations shall be in accordance with the following symbols:

Basic indicator:

IC/ - Basic indicator for Naval interior communication equipment.

Dial size:

2 - 2 inches
4 - 4 inches
6 - 6 inches

Mounting:

B - Bulkhead
P - Panel
U - Universal

Dial illumination:

R - Red
W - White
N - No illumination

Revolution counter:

C - With counter
N - Without counter

Speed range:

2 - 0 to 200 r.p.m.
4 - 0 to 400 r.p.m.

1.2.2.1 Indicator-transmitter. - The indicator-transmitter, unless otherwise specified (see 6.1), shall be of the type designated in the following form:

IC/6PNC4-Z
Separator:

A dash (-) shall be used to separate the major or basic features from the special features.

Special features:

Special features shall include the number of scales on the dial and number of pointers, as follows (and such other special features as may be specified (see 6.1):

Symbol	Feature
X	Single scale and pointer
Y	Single scale and two concentric pointers
Z	Two concentric scales and pointers

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-A-601 - Aluminum Alloy Sand Castings.

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MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power, and Pulse).
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements For.
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-E-17362 - Electronic Repair Parts Requirements, Procedures for Stock Numbering and Assignment of Government Nomenclature.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-E-22843 - Equipment, Low Noise Level, (Naval Shipboard Use).

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-710 - Synchros, 60 and 400 Cycle.

DRAWINGS

BUREAU OF SHIPS

3128837 - Indicator, Type IC/6PNC4-X.
3244956 - Indicator, Type IC/6PNC4-X.
3295478 - Indicator, Type IC/4PRN2-X, 4-X.
3295479 - Indicator, Type IC/PNN2-X, 4-X.
3295480 - List of Onboard Repair Parts.
3301753 - Transmitter, Type IC/3A.
3301754 - Transmitter, Type IC/2B.
3301758 - Transmitter, Type IC/2A.
3315293 - Speed Signal Switch.
3328618 - Indicator, Type IC/6PNC2-X, 4-X.
3331227 - Transmitter, Drive Gears.
3331228 - Transmitter, Type IC/3B.
3331229 - Indicator-Transmitter, Type IC/6PNC2-Z.
3331230 - Indicator-Transmitter, Type IC/6PNC4-Z.
3331231 - Indicator-Transmitter, Type IC/6PNC4-ZM.
3331232 - Indicator, Type IC/6PNN2-X, 4-X.
3331233 - Indicator, Type IC/4URN2-Y, 4-Y.
3331234 - Indicator, Type IC/4PNN2-X, 4-X.
9000-S6595-73687 - Standard Dial Markings for Interior Communications and Industry Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
LT-1 - Industrial Laminated Thermosetting Products.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York 17, N.Y.)

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AMERICAN SOCIETY FOR TESTING MATERIALS
B26 - Aluminum-Base Alloy Sand Castings

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

AMERICAN STANDARDS ASSOCIATION
B5.20 - Machine Pins
B17f - Woodruff, Keys, Keyslots, and Cutters
B18.1 - Small Solid Rivets
B18.2 - Square and Hexigon Bolts and Nuts
B18.3 - Socket Head Cap, Screws and Socket Set Screws
B18.6 - Slotted and Recessed Head Screws, Machine and Tapping Types
B18.6.2 - Hexagon Head Cap Screws, Slotted Head Cap Screws, Square Head Set Screws and Slotted Headless Set Screws
B18.6.4 - Slotted and Recessed Head Tapping Screws and Metallic Drive Screws
B27.1 - Lock Washers
B27.2 - Plain Washers

(Application for copies should be addressed to the American Standards Association, 70 East 54th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Qualification. - The equipment furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General requirements. - Each ship equipment shall be designed to indicate the speed and count the revolutions of the propeller shafts, and to indicate their direction of rotation. A physically driven transmitter, one for each propeller shaft, shall transmit to its associated indicator-transmitter, which in turn shall indicate propeller speed, direction, and totalized revolution count. The indicator-transmitter, in turn, shall transmit to one or more indicators, which shall merely repeat the desired data at other remote locations, as required.

3.2.1 The equipment shall be sturdily constructed of a grade of material adequate for each specific application consistent with the requirements noted herein.

3.2.2 The design, layout, and assembly of the unit and their parts shall be such as to facilitate quantity production. Jigs shall be used to position parts during welding and assembly operation.

3.2.3 Mounting dimensions shall be in accordance with the drawings referenced herein.

3.3 General features. - The equipment shall be in accordance with the following paragraphs of Specification MIL-I-983 in addition to the requirements specified herein: (Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General information
Definitions
Materials, general
Stressed or vital parts
Substitution of (equal or superior) materials
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Aluminum
Magnesium

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Iron or steel
Brass
Bronze
Copper
Dial sizes
Locking devices
Washers
Ball bearings
Parts-electrical-general
Capacitors
Relays
Synchros (see 3.14.4.4.1, 3.14.5.4 and 3.15)
Dial illumination lamps
Switches
Indicator lights and lampholders
Metallic rectifiers
Enclosures-general
Enclosure-accessibility
Enclosure-degree of (see 3.14.4.1, 3.14.5.1 and 3.14.6.1)
Enclosure-mounting
Stiffening grooves
Through bolting
Stuffing tubes
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity (special cold limit applies)
Copper-beryllium alloy
Copper-nickel alloy
Nickelcopper alloy (monel)
Copper-nickel-zinc alloy (nickel silver)
Zinc
Other metals
Springs
Plastics
Ceramics
Impregnating, filling, potting, embedding and encapsulating compounds
Glass
Lubricants and lubrication
Dials and pointers
Salt spray
Primary power supply circuits
Power supply tolerances
Personnel protection
Shielding and radio frequency noise reduction
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards, connectors and terminals
Synchro connections and markings
Wiring
Color coding
Dial illumination
Electrical insulation
Dielectric strength and insulation resistance clearances
Airborne audible noise (see 3.13)
Structureborne noise (vibration) for submarine equipment (see 3.13)
Interchangeability and standardization

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Technical manuals
Designation and marking (serial numbers required)
Design for production
Reports
Nomenclature
Workmanship and general examination

3.4 Accelerated life. - The units shall satisfactorily withstand the accelerated life test (endurance) specified in 4.5.4.

3.5 Shock, vibration and inclination. - The units shall satisfactorily withstand without mechanical or electrical damage which would cause malfunctioning or inoperation method 207 shock in accordance with Standard MIL-STD-202, vibration tests (vital), type I in accordance with Standard MIL-STD-167, or inclination tests of 60 degrees from vertical in any direction, that is, forward, backward, left or right.

3.6 Corrosion-resisting steel hardware items. - External hardware items such as nuts, screws, bolts and lockwashers shall be of nickel-copper alloy.

3.7 Screws, bolts, nuts, washers, pins and other hardware items. - All screws, bolts, nuts, washers, pins and other hardware items shall be in accordance with Publications American Standards Association (ASA) B5.20, B17f, B18.1, B18.2, B18.3, B18.6, B18.6.2, B18.6.4, B27.1 and B27.2, as applicable.

3.8 Magnetic permeability of components. - Except for those applications where the function of the component requires the use of magnetic material, all components shall use non-magnetic type materials.

3.9 Tools. - No special tools shall be necessary for assembly. disassembly or maintenance of the equipment referenced herein. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.)

3.10 Transformers. - Transformers shall be in accordance with grade 5, class S of Specification MIL-T-27, and shall have a life expectancy X.

3.10.1 Insulating electrical windings. - Electrical windings shall be thoroughly treated or impregnated by a method which will ensure the evacuation of all air and water from, and the filling in of all interstices within such windings. The liquid insulation used shall have such characteristics and be so applied as to ensure thorough drying, solidification, or curing throughout the innermost recesses of the windings.

3.11 Electrical tapes. - Electrical tapes of the best commercial grade of acetate film; glassfiber or polyethylene only shall be used. Fabric or textile pressure-sensitive (adhesive or friction) tape shall not be used.

3.12 Painting. - Sheet metal and case enclosures shall be completely painted as specified in 3.12.1 through 3.12.4.

3.12.1 Cleaning. - After all machinery, welding and brazing operations are completed, the interior and exterior surface of all metal enclosures shall have all rust or other visible corrosion products removed, and shall be thoroughly cleaned of all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing.

3.12.2 Undercoat and finish. - One coat of zinc chromate primer followed by two coats of high grade semi-gloss light gray equipment enamel shall be applied. The anodizing, primer, and enamel coats shall be applied in accordance with the best commercial practice.

3.12.3 Protection against corrosion. - All metal parts shall be of suitable corrosion-resisting materials or other materials treated in a satisfactory manner.

3.12.4 Corrosion-resisting treatments. - The corrosion-resisting treatments shall be in accordance with those specified herein. In each case the treatment shall be accomplished in accordance with the best commercial practices.

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(a) Zinc coating (hot dip galvanizing) for parts other than threaded fasteners, unless otherwise specified herein.
(b) Electroplating of zinc for surfaces not to be painted and for externally threaded parts.
(c) Electroplating cadmium for surfaces not to be painted and for externally threaded parts.
(d) Electroplating of chromium for general application.
(e) Anodic treatment of aluminum.
(f) Lubricant and lubrication. With the exception of the synchro bearings and main shaft bearings, all other bearings in the equipment shall be lubricated with light oil such as Navy 2110. Gear teeth shall be lubricated with a 600 W oil or Navy type 5190 oil.

3.13 Airborne and structureborne noise.- Airborne and structureborne noise of equipment being furnished for installation in surface ships shall meet the requirements of Specification MIL-I-983 and equipments furnished for installation in submarines or for submarine stock support shall meet the grade B noise requirements of Specification MIL-E-22843, as specified (see 6.1).

3.14 Detail requirements.-

3.14.1 Counters.- All revolution counters shall be of the odometer, nonreset, ball bearing, revolving drum type with six figure wheels. The numerals on the figure wheels shall be at least 1/4 inch high by 3/16 inch wide. The numerals shall be white on a dull black background.

3.14.2 Reverse direction indicators.- Reverse direction indicators shall consist of a lamp type (red) signal, which shall, when energized, indicate the reverse (astern) rotation of the propeller shafts. The signal shall be continuous during reverse rotation of the propeller shaft and shall utilize a 6 to 8 volt lamp. Lamps and lamp holders shall be in accordance with Specification MIL-I-983.

3.14.3 Accuracy.- When the equipment, consisting of a transmitter, indicator-transmitter, and three indicators connected in a system, is operated on designed voltage and frequency at any ambient temperature over the range 40 to 130°F., the accuracy shall be as specified in 3.14.3.1 and 3.14.3.2.

3.14.3.1 Permissible error.- Maximum permissible error of r.p.m. indication in the indicator-transmitter and indicators shall be as specified in (a) and (b).

(a) Indicator-transmitter.-
From 5 r.p.m. to 1/10 of full scale reading, error shall be within ±1/2 of 1 percent of full scale reading. From 1/10 of full scale reading to full scale reading, error shall be within ±1/4 of 1 percent of full scale reading.
(b) Indicators.-
(1) Single pointer indicators.- Error shall not exceed a tolerance of ±1/2 of 1 percent of full scale reading.
(2) Multiple pointer indicators.- Error shall not exceed a tolerance of ±3/4 of 1 percent of full scale reading.

3.14.3.2 Response time.- The accuracy requirements of 3.14.3.1 shall be met within 5 seconds after the propeller shaft assumes constant speed. The indicator-transmitter shall also be capable of indicating from 5 r.p.m. to full scale within 1 minute.

3.14.4 Transmitter.- The transmitter shall provide a means for transmitting synchro signals indicative of propeller shaft speed to the associated indicator-transmitter. The transmitter shall contain in a single enclosure a synchro driven by a shaft which is geared or directly coupled as specified in 1.2.1 to the propeller shaft or propulsion machinery of the ship. The transmitter shall also contain a total-revolution counter as specified in 3.14.1 and contacts to operate reverse-direction signals as specified in 3.14.2 and 3.14.4.6. Transmitter types shall conform to Drawings 3301758, 3301753, 3301754 and 3331228.

3.14.4.1 Enclosure.- The enclosure shall be fabricated of 356 T-6 cast aluminum alloy conforming to Publication American Society for Testing Materials (ASTM) B26 or class 3M, T-6 cast aluminum in accordance with Specification QQ-A-601. The enclosure shall be watertight (3-foot), and shall be designed for deck mounting with four mounting feet.

3.14.4.2 Dimensions and weights.- Unless otherwise specified in the contract or order, the transmitter, including mounting feet and shaft extension, but exclusive of terminal tubes, shall not exceed the dimensions or weights specified in table I.

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Table I - Dimensions and Weights

Length (inches)	Width (inches)	Depth (inches)	Weight (pounds)
15	12	6	40
12	12	9	30

3.14.4.3 Operating speed.- The maximum operating speed shall be as required for the applicable type (see 1.2.2)

3.14.4.4 Transmitter mechanism.-

3.14.4.4.1 Synchro.- A type 37TX6 synchro, conforming to Standard MIL-STD-710 shall be incorporated in the transmitter for transmission to the associated indicator-transmitter. The transmitting synchro shall be driven at one-half the speed of the propeller shaft. A unidirectional device shall be incorporated in the transmitter so that the transmitting synchro shall be driven in the "ahead" direction regardless of the direction of rotation of the propeller shaft. One revolution of the propeller shaft shall be sufficient to operate the unidirectional device.

3.14.4.4.2 Contacts.- The transmitting mechanism shall close electric contacts for operating reverse direction signals (see 3.14.2) in the associated indicators when the propeller shaft rotates in the "astern" direction. One revolution of the propeller shaft shall be sufficient to completely operate the unidirectional device and the reverse direction signal contacts.

3.14.4.4.3 Counter.- A counter (see 3.14.1) shall be mechanically driven from the transmitter drive shaft, and shall count the revolutions made by the propeller shaft regardless of its direction of rotation. In operation of the unidirectional device, not more than 1/4 of a revolution in the counter shall be lost.

3.14.4.4.4 Accessibility.- The transmitter shall be so designed that the transmitting mechanism may be removed from the enclosure for servicing without disturbing the operation of the total revolution counter. The design shall also be such as to minimize maintenance and upkeep. Means shall be provided to permit the necessary lubrication of parts without disassembly of the transmitter.

3.14.4.4.5 Transmitter drive shaft.- The transmitter drive shaft shall project 2 inches from the enclosure and shall be 5/8-inch in diameter. The design of the transmitter shall be such that the drive shaft can be readily transposed from one side of the enclosure to the other in order to provide flexibility of installation. The transmitter drive shaft shall be driven as specified in 1.2.1 either by gears from the propeller shaft or by means of a flexible coupling connected directly to a stub shaft on the main motor or propelling machinery. When gear drive is required, the transmitter drive shaft shall be threaded and provided with a lockwasher, nut, and key to secure the shaft gear. For the flexible coupling or direct drive, no modification to the transmitter shaft is required, unless specified in the contract or order.

3.14.4.4.6 Gear drive.- The gear drive assembly shall consist of one propeller shaft gear (driving gear) and one transmitter gear (driven gear). The drive shaft bearings shall be sealed for life type ball bearings. The propeller shaft gear (driving gear) shall be made of bronze, and shall be of the split ring type. The transmitter shaft gear (driven gear) shall consist of a bronze hub and a plastic outer rim in which teeth are cut in accordance with Drawing 3331227. Plastic material used in this gear shall be grade C in accordance with Publication National Electrical Manufacturer's Association LT-1. The gears shall conform to the following design:

Diametrical pitch - 6
Type of teeth - Involute
Pressure angle - 14 - 1/2 degrees
Shaft gear face:
  Propeller shaft gear - 5/8 inch
  Transmitter shaft gear - 1/2 inch
Ratio between shaft-gear and transmitter gear - 3 to 1 (unless otherwise specified in type designation, that is:

$$\frac{\text{Number of teeth on driving gear}}{\text{Number of teeth on driven gear}} = 3)$$

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3.14.4.4.6.1 Bore.- Shaft gears shall be rough-bored to diameters as specified (see 6.1). Final bore will be accomplished by the installing activity. The propeller shaft gears shall be so designed that the rough-bore diameter may be increased by the installing activity to a final bore of 2 inches greater than the rough-bore diameter.

3.14.4.4.7 Direct drive.- The flexible coupling shall be furnished by the installing activity.

3.14.5 Indicator-transmitter.- The indicator-transmitter shall consist of a single enclosure containing the mechanism necessary to receive the output of the transmitter specified in 3.14.4 to perform the functions specified in (a) through (d).

(a) Indicate propeller shaft speed in r.p.m.
(b) Indicate total propeller shaft revolutions.
(c) Indicate reverse direction of rotation of the propeller shaft.
(d) Transmit propeller shaft speed in r.p.m. to remote indicators.

The indicator-transmitters shall conform to the respective types shown on Drawings 3331229, 3331230 and 3331231.

3.14.5.1 Enclosure.- The enclosure shall be fabricated of 356 T-6 cast aluminum alloy conforming to Publication ASTM B26, alloy SG-70A, T-6 or class 3M, T-6 cast aluminum conforming to Specification QQ-A-601. The enclosure shall be dripproof. The mounting shall be as required by the applicable type (see 1.2.2).

3.14.5.2 Operating speed range.- The operating speed range shall be as required by the applicable type (see 1.2.2).

3.14.5.3 Indicator-transmitter mechanism.- The indicator-transmitter mechanism shall continuously indicate the actual r.p.m. and instantly adjust r.p.m. readings to any change in the propeller shaft speed over the range of speed from 5 r.p.m. to full scale, and shall conform to the accuracy requirement specified in 3.14.3. Suitable limit switches or mechanisms shall be incorporated to prevent injury or excessive wear at overspeeds or at zero speed. The overspeed limiting device shall prevent jamming or damage to the mechanism by speeds up to 115 percent full scale reading. The low speed limiting device shall prevent excessive wear of the mechanism by stopping all constant speed mechanism, such as friction disks, constant speed motor, and so forth, when the system is energized and the shaft speed is zero. The speed signal switch shall be in accordance with Drawing 3315293.

3.14.5.4 Synchros.- The mechanism shall incorporate a type 31TR6 synchro unit in accordance with Standard MIL-STD-710 to receive the synchro signal from the transmitter, and shall convert this signal into an indication of propeller shaft speed. The mechanism shall also include a type 37TX6 synchro unit in accordance with Standard MIL-STD-710 for transmitting speed indications to remote indicators. One revolution (360 degrees) of the type 37TX6 transmitting synchro shall represent full scale deflection at the indicator.

3.14.5.5 Counter.- When counters are required by the contract or order, the counter as specified in 3.14.1 shall be incorporated in the mechanism and mechanically driven from the type 31TR6 synchro specified in 3.14.5.4 to indicate the total revolutions made by the propeller shaft regardless of its direction of rotation.

3.14.5.6 Reverse direction indicator.- A reverse-direction indicator as specified in 3.14.2 shall be provided.

3.14.5.7 Dial.- The size of dial and the number of scales and pointers shall be as specified in 1.2.2 for the applicable type designation. The dial shall be calibrated in accordance with Drawing 9000-S6595-73687. The dial shall also include the reverse-direction indicator signal and the total-revolution counter. Dial illumination shall be provided when required by the applicable type (see 1.2.2).

3.14.5.8 Size and weight.- The indicator-transmitter shall be as small and lightweight as practicable commensurate with the requirements of this specification. Special emphasis shall be given in the design toward a reduction in size and weight of parts, assemblies, and structural members. Unless otherwise

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specified in the contract or order, the indicator-transmitter, including mounting feet and exclusive of terminal tubes, shall not exceed the dimensions or weights specified in table II.

Table II - Dimensions and weights.

Length (inches)	Width (inches)	Depth (inches)	Weight (pounds)
12	12	10	40

3.14.5.9 Interchangeability.- The design shall incorporate an arrangement whereby the speed range of the indicator-transmitter can be readily changed (that is: 200 to 400, and so forth) by means of interchangeable dials and associated gears. The arrangement shall be such that this interchangeability can be accomplished aboard ship without requiring recalibration of the indicator-transmitter.

3.14.5.9.1 The indicators shall conform to the respective types shown on Drawings 3244956, 3295478, 3295479, 3331232, 3331233, 3128837, 3328618 and 3331234.

3.14.6 Indicators.- The indicator shall contain, in a single enclosure, the mechanism necessary to receive the output from the indicator-transmitter and indicate the information specified in (a) and (b).

(a) R.p.m. of the associated propeller shaft.
(b) Reverse direction of rotation of the associated propeller shaft.

3.14.6.1 Enclosure.- The enclosure shall be fabricated of 356 T-6 cast aluminum alloy conforming to Publication ASTM B26, alloy SG-70A, T-6 or class 3M, T-6 cast aluminum in accordance with Specification QQ-A-601. The enclosure shall be watertight (3-foot). The mounting shall be as required by the applicable type (see 1.2.2).

3.14.6.2 Dimensions and weights.- Unless otherwise specified in the contract or order, the indicators, including mounting feet and exclusive of terminal tubes shall not exceed the dimensions and weights specified in table III.

Table III - Dimensions and weights

Dial size	Length (inches)	Width (inches)	Depth (inches)	Weight (pounds)
2 and 4 inch	6	6	6	15
6 inch	10	10	8	20

3.14.6.3 Operating speed range.- The operating speed range shall be as required by the applicable type (see 1.2.2).

3.14.6.4 Operating indicator.- The indicator shall operate satisfactorily when tested as specified in 4.5.1.

3.15 Indicator mechanism.- The indicator mechanism shall incorporate synchro units in accordance with Standard MIL-STD-710 as specified in (a) and (b).

(a) Indicators with 4 inch dials - type 23TR6 synchro.
(b) Indicators with 6 inch dials and up - type 31TR6 synchro.

3.15.1 Counters.- When counters are required by the contract or order, the counter specified in 3.14.1 shall be incorporated in the mechanism and shall be mechanically driven from a type 23TR6 synchro to indicate the total revolutions made by the propeller shaft regardless of its direction of rotation.

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3.15.2 Dial. - The size of the dial and the number of scales and pointers shall be as specified in the type designation specified in 1.2.2. The dial shall be calibrated in accordance with Drawing 9000-S6595-73687. The dial shall also include the reverse direction indicator specified in 3.14.2. Dial illumination shall be provided when required by the applicable type (see 1.2.2).

3.16 Subassemblies. - Subassemblies using gears and levers shall be secured to their associated shafts by positive locking action employing either pins, keys, nuts and so forth. Set screws shall not be used as the principal method of securing these components.

3.17 Drawings. -

3.17.1 Drawing requirements. - The drawings and features applicable to each type shall consist of a schematic diagram, a wiring diagram, and assembly drawings.

3.17.1.1 Appropriate notes shall be added to the drawings giving the information specified in (a) and (b).

(a) Power requirements under "full load" and "standby" conditions.
(b) Heat dissipation under "full load" and "standby" conditions.

3.17.1.2 Wiring diagrams shall include the information specified in (a) through (e).

(a) All parts in correct relation as to physical location, with terminals clearly shown and marked and all wiring between parts.
(b) The coding of all wires.
(c) Appropriate notes shall designate wire size and type.
(d) Designation of parts shall correspond to those assigned in the diagrams.
(e) Voltage and resistance readings at key service points.

3.17.2 Commencement of work by manufacturer. - Upon receipt of drawing approval letter and prior to validation of tracings, and, if the manufacturer has no objections to any changes or modifications requested in the drawing approval letter, manufacture may be started in accordance with the drawings as revised in anticipation of validation of drawings.

3.17.3 Approval of drawings by the bureau or agency concerned does not relieve the manufacturer of the responsibility for meeting all the provisions of the contract.

3.17.4 Drawings for provisioning. - Drawings required solely for identification and cataloging purposes shall be submitted in accordance with Specification MIL-P-15137 or Specification MIL-E-17362 as applicable.

3.18 Onboard repair parts. - Onboard repair parts shall be furnished in accordance with Drawing 3295480 as specified (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests[1/]. - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of those tests specified in 4.2.1.

4.2.1 Sample for qualification tests. - One equipment of each type shall be subjected to the tests listed below and the tests shall be performed, in general, in the order listed.

[1/] Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.2.1).

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Test	Requirement paragraph	Test paragraph
General examination	3.3	MIL-I-983
Supply line voltage and frequency	3.3	MIL-I-983
Radio frequency noise interference	3.3	MIL-I-983
Airborne (audible noise)	3.3, 3.13	MIL-I-983
Airborne and structure borne noise	3.3, 3.13 (as applicable)	4.4.2 (as applicable)
Inclination	3.5	4.4.3
Accelerated life	3.4	4.4.4
Dielectric strength	3.3	MIL-I-983
Insulation resistance	3.3	MIL-I-983
Operating test	3.14.6.4	4.4.1
Accuracy	3.14.3 3.14.3.1 3.14.3.2	4.4.5
Enclosure	3.3, 3.14.4.1 3.14.5.1 3.14.6.1	4.4.6
Vibration	3.5	MIL-STD-167
Shock	3.5	MIL-STD-202 (method 207)
Final accuracy	3.14.3 3.14.3.1 3.14.3.2	4.4.5

4.3 Acceptance inspection. -

4.3.1 Group A tests. - Each equipment on the contract or order shall be subjected to the tests shown in table IV.

Table IV - Group A tests.

Test	Requirement paragraph	Test paragraph
General examination	3.3	MIL-I-983
Dielectric strength	3.3	MIL-I-983
Insulation resistance	3.3	MIL-I-983
Operating test	3.14.6.4	4.4.1

4.3.2 Group B tests. - Group B tests shall be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to each of the group B tests shown in table V. The results of each test shall be compared with the specification for any group B test. The contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

Table V - Group B tests.

Test	Requirement paragraph	Test paragraph
Supply line voltage and frequency variation	3.3	MIL-I-983
Airborne and structure borne noise	3.3, 3.13 (as applicable)	4.4.2 (as applicable)
Radio frequency noise interference	3.3	MIL-I-983
Inclination	3.5	4.4.3
Accelerated life	3.4	4.4.4

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Table V - Group B tests. (Cont'd)

Test	Requirement paragraph	Test paragraph
Accuracy	3.14.3, 3.14.3.1 3.14.3.2	4.4.5
Enclosure	3.14.5.1, 3.14.6.1	4.4.6
Vibration	3.5	MIL-STD-167
Shock	3.5	MIL-STD-202 (method 207)
Final accuracy	3.14.3, 3.14.3.1 3.14.3.2	4.4.5

4.4 Test procedures. -

4.4.1 Operating test. - The operating test shall consist of one hour operating periods with a transmitter driving and indicator transmitter and one or more indicators. The transmitter shall be driven at speeds varying from approximately 10 r.p.m. to full scale reading in both forward and astern directions.

4.4.2 Airborne and structureborne noise test. - Compliance with the airborne (audible noise) requirements of 3.13 shall be determined in accordance with the requirements of Specification MIL-I-983 for equipment being furnished for surface ships. Compliance with the airborne and structureborne noise requirements shall be determined in accordance with the requirements of Specification MIL-E-22843 for all equipment being furnished for submarines or for submarine stock support.

4.4.3 Inclination test. - The units shall operate satisfactorily when inclined to 60 degrees as specified in 3.5. When conditions permit, this test may be performed during the accelerated life test.

4.4.4 Accelerated life test. - The accelerated life test shall be conducted for a period of 500 hours. The 500-hour test shall be divided approximately as shown in table VI.

Table VI - Accelerated life test.

Hours	Ambient temperature during test	Dial indications	Angle of tilt
0 to 50	Conduct initial accuracy test 1/ 75°F. Nominal (room temp.)	Constantly varying between 5 r.p.m. to full scale reading	Normal
50 to 175	50°F. at approximately 90 percent relative humidity for at least 50 percent of period.	Constantly varying between 5 r.p.m. to full scale reading	Normal
175 to 300	130°F. at approximately 90 percent relative humidity for at least 50 percent of period	Constantly varying 2/ between 5 r.p.m. to full scale reading	Normal
300 to 308	75°F. Nominal (room temp.)	Constantly varying between 5 r.p.m. to full scale reading	Incline 60 degrees forward from vertical
308 to 316	75°F. Nominal (room temp.)	Constantly varying between 5 r.p.m. to full scale reading	Incline 60 degrees back from vertical

See footnotes at end of table.

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Table VI - Accelerated life test. (Cont'd.)

Hours	Ambient temperature during test	Dial indications	Angle of tilt
316 to 324	75°F. Nominal (room temp.)	Constantly varying between 5 r.p.m. to full scale reading	Incline 60 degrees from vertical
324 to 332	75°F. Nominal (room temp.)	Constantly varying between 5 r.p.m. to full scale reading	Incline 60 degrees right from vertical
332 to 500	75°F. Nominal (room temp.)	Full scale	Normal

1/ At end of 50 hours, conduct accuracy check test, vibration test, and short test runs (1/2 hour each) at undervoltage and underfrequency and at overvoltage and overfrequency.

2/ A start and stop condition that will duplicate actual in-service operation shall be introduced for this phase of the test. A minimum of twenty "starts-stops" at the end of each hour's test is required. The deenergized period shall be of sufficient duration so as to permit the equipment under test to come to a complete rest. The rest period shall not exceed thirty seconds.

4.4.5 Accuracy test. - Accuracy tests shall be conducted at ambient temperatures of 40°, 75°, and 130°F. The accuracy shall be checked alternately in the "head" and "astern" direction at 5, 10, and each multiple of 10 r.p.m. to full scale of components having a range of 200 r.p.m. at 5, 15, and each multiple of 15 r.p.m. to full scale for components having a range of 400 r.p.m. and up. An accuracy check test shall also be made at a speed of 10 percent above the full scale reading. Accuracy shall conform to 3.14.3.1.

4.4.6 Enclosure test. - A thorough inspection shall be made of the enclosure to determine conformance with the requirements specified herein. There shall be no leakage of water into the enclosure or leakage of air from the enclosure under test when the enclosure is subjected to the submergence test shown on figure 1.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Propeller revolution indicating equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation (180 days).

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

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5.1.2 Onboard repair parts. - The onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively, in accordance with Specification MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment requirements. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging, packing and marking. - The equipment and accessories, repair parts, and technical publications shall be preserved and packaged levels A or C; packed level A or B as specified, and marked in accordance with Specification MIL-E-17555.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of specification.
(b) Type and number of transmitters, indicator-transmitters and indicators required (see 1.2).
(c) Any additional special features required (see 1.2.2).
(d) Whether enclosures shall be designed for universal bulkhead, pedestal, panel, console, or special mountings (see 1.2.1 and 1.2.2).
(e) Airborne and structureborne noise reduction requirements required (see 3.13).
(f) Shaft gears diameter required (see 3.14.4.4.6.1).
(g) Onboard repair parts required (see 3.18).
(h) Selection of applicable level of preservation, packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 15554, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2.1).

6.2.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacturer, use, or sell any patented invention that may in any way be related thereto.

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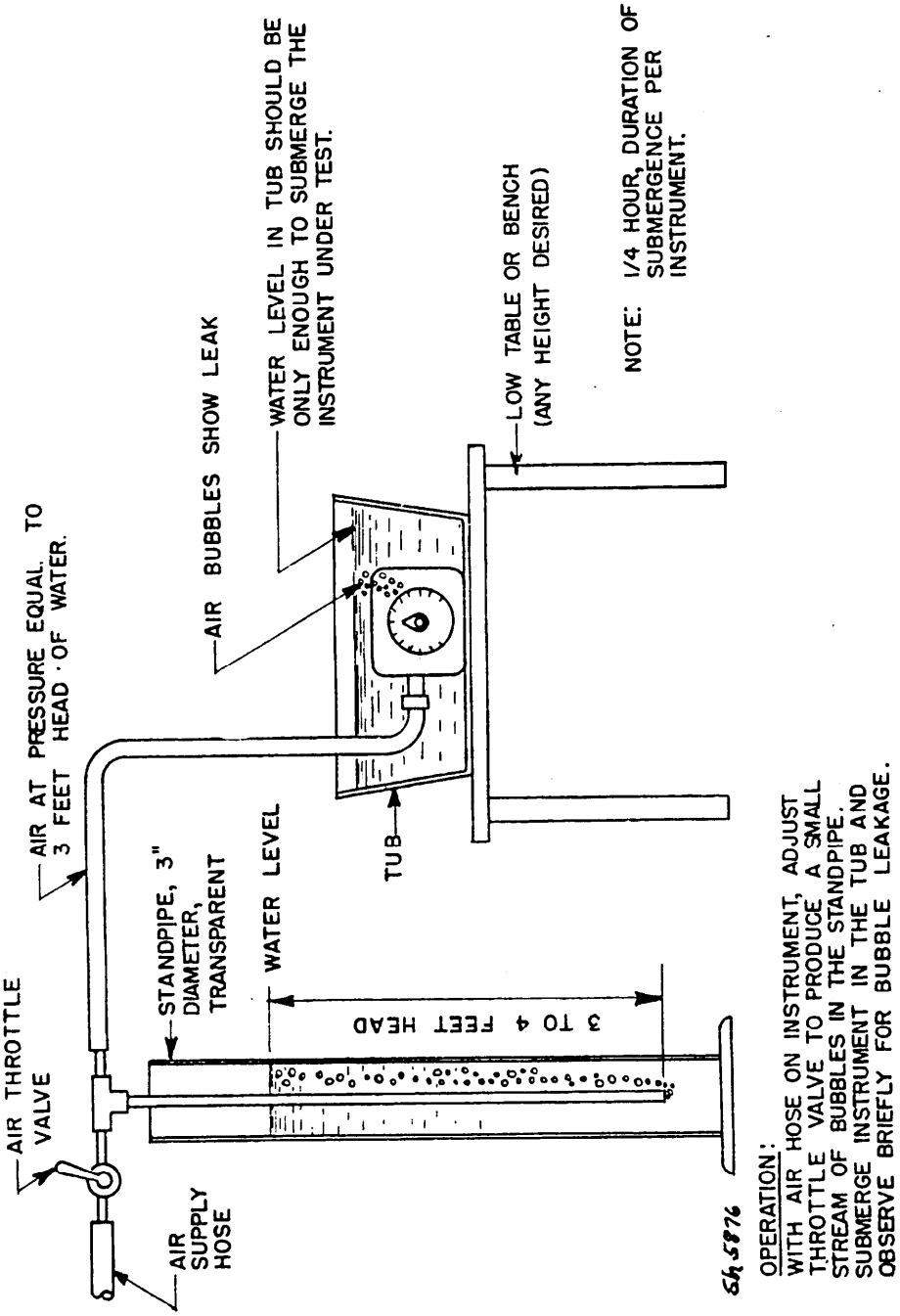


Figure 1 - Submergence test apparatus.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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